

INTRODUCTION TO NETWORKS

Basic Network Concepts

- A computer **network** is a system of computers, network devices, printers, and software linked by communications cabling.
- The **Internet** is a global network of thousands of smaller networks linked by various network equipment and communication methods.
- The **World Wide Web (Web)** is a vast network of servers worldwide that provide access to voice, text, video, and data files.

Network Communication Methods

1. Wire
2. Fiber-optic communications
3. Radiowaves such as microwaves

Communications Media and Nodes

- A **node** is any device connected to a network, such as a microcomputer, a mainframe, a minicomputer, network equipment, or a printer.
- **Communications media** are the cabling or radio waves used to connect one network computer to another and to transport data between them.
- **Spread spectrum technology (SST)** is used by wireless networks in place of cable for communications between network nodes. Network data is transmitted by means of reliable high-frequency radio signals.

Critical Types of Network Nodes

1. Workstation
2. Hosts
3. Servers

Workstations, Hosts, and Servers

- A **client** computer accesses a mainframe, a minicomputer, or a microcomputer that allows access to multiple users.
- A **host** has an operating system that allows multiple computers (clients) to access it simultaneously.
- A **workstation** has its CPU and may be used as a stand-alone computer for local software applications.
- A **file server** is a single network computer that offers multi-user access by making software applications, data files, and network utilities available to other network computers.

A Network with Many Types of Nodes

1. **Hub** - A **network hub** is a node that broadcasts data to every computer or Ethernet-based device connected to it.
2. **Switch** - A **network switch** forwards data between devices, unlike routers, which forward data between networks
3. **Router** - A **router** is a device that connects two or more packet-switched networks or subnetworks.
 - a. Serves two primary functions:
 - i. Managing network traffic
 - ii. Connection sharing

Network Interface Card (NIC)

- A **Network Interface Card (NIC)** is an adapter card that enables a workstation, file server, printer, or other device to connect to a network and communicate with other network nodes.

Packets, Frames, and Cells

- A **packet** is a data unit formatted for transmission over a network.
- **Frame** refers to the packet format or contents and routing information.
- A **protocol** is an established guideline that specifies how networked data is formatted

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into a packet, how the packet is transmitted, and how it is interpreted at the receiving end.

Functions of Network Communications Devices

- Connect different nodes into one or more networks
- Amplify packet signals to go further
- Connect different networks into one or more networks
- Control the packet traffic to reduce network congestion
- Route packets and cells to particular destinations
- Establish alternative routes for packets when portions of the network are down
- Connect networks within a city or between continents
- Monitor for network problems

Networking Capabilities

- File services
- Print services
- Software services
- Client/server applications
- Electronic mail services
- Internet and intranet services
- Commerce
- Network monitoring

Network Sharing Services

- Sharing files is a hallmark of networks. A single user can make certain files, peripherals, and other software available to a limited set of users.

Internet and Intranet Services

- The **Internet** is a prime example of networking capabilities.
- An **Intranet** is a private network within an organization. It uses the same Web-based software as the Internet, but its access is restricted.

- **Hypertext Markup Language (HTML)** is a formatting process that enables documents and graphics images to be read on the Web. It also provides for fast links to other locations.
- A **Web browser** enables the user to search for information and to display text, graphics, sound, and video from the World Wide Web.

Peer-to-Peer Networking

- A **peer-to-peer network** is one in which any computer can communicate with other networked computers on an equal or peer-like basis without going through an intermediary (server or host); it is one of the simplest ways to network and is effective for very small networks.

Workgroup

- A **workgroup** consists of users who share resources (such as drive and printer resources) in a decentralized way.

Potential Downsides of Peer-to-Peer Networking

- Offers only moderate network security
- As the number of network users grows, so does the need for a central place to store and manage information.
- Can experience slow response because the operating systems are not optimized for heavy multiple access to one computer

Wired Networking

- Traditional way to build a network
- Workstations, servers, hosts, and network equipment are physically connected to the cable through connectors, NICs, and other physical connection devices.

Wireless Networking

- An alternative to communications cable, this method links networks at two locations

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using radio communications; it is often cheaper and easier to install than cable, remains reliable, and is difficult for intruders to intercept due to its use of ultra-high frequency ranges.

LANs, MANs, WANs, and Enterprise Networks

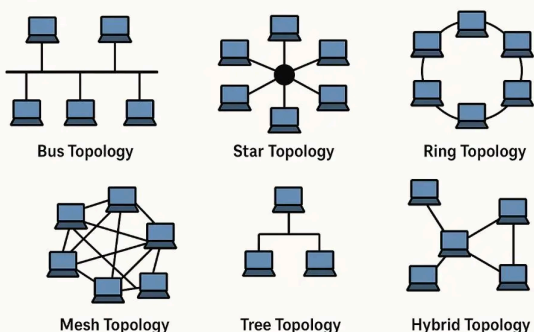
- A **local area network (LAN)** is a series of interconnected computers, printers, and other computer equipment that share hardware and software resources.
- A **metropolitan area network (MAN)** links multiple LANs within a large city or metropolitan region.
- An **enterprise network** reaches throughout a large area (such as a college campus, a city, or across several states) connecting many kinds of LANs and network resources.
- A **wide area network (WAN)** is a far-reaching system of networks. WANs can extend across states and across continents

TOPOLOGIES, COMMUNICATIONS MEDIA, AND IP ADDRESSING

Network Topologies

- **Two Components:**
 - Physical layout of the cable
 - Logical path followed by network packets sent on the cable

MOST COMMON TYPES OF NETWORK TOPOLOGY



- **Main Topologies:**
 1. Bus, Ring,
 2. Star,
 3. Mesh,
 4. Tree,
 5. Hybrid,
 6. Peer-to-Peer.
- **Cable Plant:** The total cabling of the network.
- **Network Traffic:** The number, size, and frequency of packets transmitted on the network in a given amount of time.
- **Mission-Critical:** A software application or hardware service that involves a vital business or support function.
- **Fault Tolerance:** Availability of backup hardware or software so functions continue without interruption in the event of a problem.

Bus Topology

- Nodes are connected to a segment of cable in the logical shape of a line, with a terminator at each end.
- **Advantages:** Works well for small networks; inexpensive; requires less cable; easy to add workstation.
- **Disadvantages:** Management costs high; difficult to isolate malfunctions; defective node can take down entire network; subject to congestion.

Ring Topology

- A network in the shape of a ring or circle, with nodes connected around the ring.
- **Advantages:** Easier to manage than a bus; easier to locate problems; good over long distances; handles high-volume traffic; reliable.
- **Disadvantages:** Requires more cable and equipment at start; not as many equipment options; not as widely used as bus.

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Star Topology

- Configured with a **central hub** and **individual cable** segments connected to the hub.
- Hub**: A central device that joins single cable segments or individual LANs into one network.
- Advantages**: Easier to manage and expand; easier to locate problems; well-suited for high-speed networking; more equipment options.
- Disadvantages**: Susceptible to a single point of failure; requires more cable at the start.

Bus-Star Topology

- Combines logical communications of a bus with the physical layout of a star.
- Backbone**: High-capacity communications medium that joins networks and central devices.
- Advantages**: Easier to manage/locate problems than traditional bus; many expansion options; no exposed terminators.
- Disadvantages**: Susceptible to a single point of failure (the hub); requires more cable.

Communications Media

- Thick coaxial cable
- Thin coaxial cable
- Shielded twisted-pair cable
- Unshielded twisted-pair cable
- Fiber-optic cable
- Wireless technologies

Coaxial Cable (Coax)

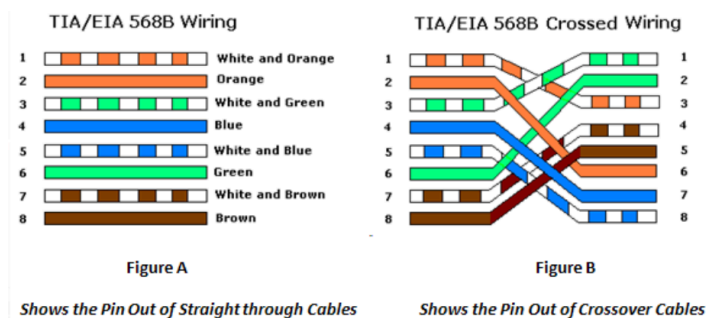
- Components**: **Copper core**, **insulation**, **conducting metal** (braided wire), **outer insulating material**.
- Varieties**: Thick and Thin (Thin uses BNC and T-connectors).
- Thin Coaxial Cable**:
 - Resembles television cable

- Attached to a **BNC connector**, which is then connected to a **T-connector**
- Middle of the **T** is connected to the **NIC** in the computer or network device
- A **T-connector** with a **terminator** at one end

- Impedance**: The total amount of opposition to the flow of current.
- Baseband**: The entire channel capacity used by one data signal.
- Broadband**: Several transmission channels on a single medium.
- Bandwidth**: Capacity of a channel to transmit data.

Twisted-Pair Cable

- Components**: **Pairs of insulated copper wires twisted together** (for reduction of Electro Magnetic Interference and RFI – Radio Frequency Interference); covered with an outer insulating jacket.
- Shielded Twisted-pair (STP)**: **Pairs of insulated wires are twisted together**, surrounded by a shielding material for added EMI and RFI protection, all inside a protective jacket.
- Unshielded Twisted-pair (UTP)**: **Most frequently used network cabling**; low cost; relatively easy installation; has no shielding material.
- Color Coding of Straight-through and Crossover Cabling**:



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Fiber-Optic Cable

- **Components:** One or more glass or plastic fiber cores inside a protective cladding material; covered by a PVC outer jacket.
- **Fat Pipe:** Fiber-optic cable used on a network backbone for high-speed communications.
- **Attenuation:** The amount of signal that is lost as the signal travels through the communications medium from its source to the receiving node.
- **Power Budget:** The difference between the transmitted power and the receiver sensitivity. It is the minimum transmitter power and receiver sensitivity needed for a signal to be sent and received intact.

Wireless Technologies

- Radio technologies
- Infrared technologies
- Microwave technologies
- RFID Technologies

Radio Technologies

- **Hertz:** The measurement of the frequency of the alternation of an electrical current or radio wave.
 - **Kilohertz (kHz)** = 1000 hertz
 - **Megahertz (MHz)** = 1 million hertz
 - **Terahertz (THz)** = 1 trillion hertz
- **Line-of-site transmission:** The signal goes from point to point.

Infrared Technologies

- Can be broadcast in a single direction or in all directions.
- Uses an **LED** to transmit and a **photodiode** to receive.

Microwave Technologies

- **Terrestrial microwave systems:** Transmit the signal between two directional antennas shaped like dishes.

- Frequency range: 4-6 GHz or 21-23 GHz.

- Requires an **FCC (Federal Communications Commission)** license.

- **Satellite microwave systems:** Transmit the signal from transmitting antenna to a satellite in space, then to the receiving antenna.

- Frequency range: 11-14 GHz.
- Requires launching a satellite or leasing service.

ADVANTAGES AND DISADVANTAGES OF WIRELESS

Technology	Advantages	Disadvantages
Radio	Inexpensive alternative where cabling is difficult; option for portable licensing.	May not be feasible for high-speed; subject to interference (military, cell phones, natural obstacles).
Infrared	Inexpensive alternative where cabling is difficult; signal difficult to intercept secretly.	May not be feasible for high-speed; subject to interference from light; does not go through walls.
Microwave	Alternative where cabling is difficult; over long distances.	May not be feasible for high-speed; expensive to install/maintain;

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ADVANTAGES AND DISADVANTAGES OF WIRELESS

subject to weather and EMI.

- Bus design
- CPU design
- Serial port design
- Parallel port design
- Keyboard architecture
- Display architecture
- Modem communications
- Printer communications
- Network communications

IP Addressing

- **IP Address:** A numerical label assigned to devices connected to a computer network that uses the IP for communication.
- **Function:** Acts as an identifier for a specific machine; helps develop a virtual connection between source and destination.
- **Format:** Consists of four numbers, each containing one to three digits, separated by a single dot (.).
- **Two Parts of an IP Address:**
 - **Prefix:** Identifies the physical network (Network Address).
 - **Suffix:** Identifies the individual computer (Host Address).

IPv4 Address Classes and Ranges						
Address Class	Type	Range	Default Subnet Mask	Number of Networks	No of Hosts Per Network	Use
A	Public	1 0 0 0 to 127 0 0 0	255.0.0.0	126	16,777,214	Governments and Large Number of Hosts
	Private	10.0.0.0 to 10.255.255.255				
B	Public	128.0.0.0 to 191.255.255.255	255.255.0.0	16,382	65,534	Medium Companies
	Private	172.16.0.0 to 172.31.255.255				
C	Public	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,150	254	Small Companies and LANs
	Private	192.168.0.0 to 192.168.255.255				
D	N/A	224.0.0.0 to 239.255.255.255	Not Applicable	N/A	N/A	Reserved for Multicasting
E	N/A	240.0.0.0 to 254.255.255.255	Not Applicable	N/A	N/A	Experimental
Special	Special	127.0.0.1 to 127.255.255.255	N/A	N/A	N/A	Loopback Testing

Standards for Networking

- Provide common ground for transmitting data, manufacturing compatible network equipment, and designing operating systems for use on a network.
- Define the maximum time a packet has to travel from one node to another.
- Define how packets are handled.

Key Networking Standards Organizations

- **American National Standards Institute (ANSI)**
- **Institute of Electrical and Electronics Engineers (IEEE):** An organization of scientists, engineers, technicians, and educators that has influenced standards for network cabling and data transmissions.
- **Consultative Committee on International Telegraph and Telephone (CCITT):** An international standards body that develops telecommunications standards for modems, digital telephone systems, and e-mail.
- **International Standards Organization (ISO):** An international body that establishes communications and networking standards; best known for contributions to network protocol standards.

The OSI Model

- The **Open Systems Interconnect (OSI)** model provides a framework for network communications based on seven functional layers.

NETWORKING STANDARDS AND MODELS

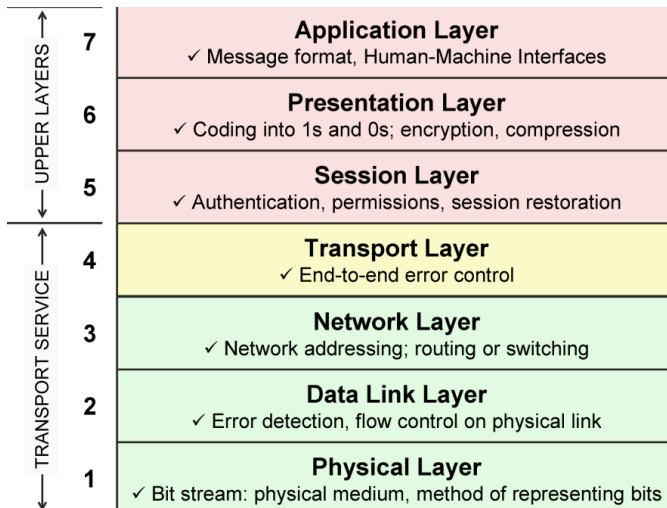
Types of Standard

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- Developed by the **ISO** and **ANSI**.
- Affects how signals are transmitted, packets are constructed, reliability, error detection, and the sharing of network resources.



Layer 1: Physical Layer

- Provides the transfer medium and translates data into transmission signals to be sent along the medium, while managing the physical layout, monitoring for errors, and determining both voltage levels for synchronization and the signal type as digital or analog.
- **Analog transmission**: One that can vary continuously, such as in a wave pattern with positive and negative voltage levels.
- **Digital transmission**: Has distinct levels to represent binary 1s and 0s, such as On and Off or +5 volts and 0 volts.
- **Interference**:
 - **Electromagnetic interference (EMI)**: caused by magnetic force fields that are generated by electrical devices.
 - **Radio frequency interference (RFI)**: caused by electrical devices that emit radio waves at the same frequency used by network signal transmission

Layer 2: Data Link Layer

- Constructs data packets in the appropriate format, creates and checks **CRC** information for error detection, manages data retransmission when errors occur, initiates the communications link for physical reliability, and examines packet addresses to acknowledge receipt.
- **Cycle redundancy check (CRC)**: An error detection method that calculates a value for the total size of the information fields contained in the frame.
- **Logical link control (LLC)**: A data link sublayer that initiates the communication link between nodes and ensures that the link is not broken unintentionally.
 - **Type 1 operation (connectionless service)**: does not establish logical connection between sending and receiving nodes.
 - **Type 2 operation (connection-oriented service)**: logical connection must be established before full communications begin.
- **Media access control (MAC)**: A data link sublayer that examines addressing information contained in network frames and controls how devices share communications on the same network.

Layer 3: Network Layer

- Determines the network path for routing frames, helps reduce network congestion, establishes virtual circuits, and routes frames to other networks while resequencing packet transmissions as needed.
- **Router**: A physical device that contains software to enable frames formatted on one network to reach a different network in a format that the second network understands.

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- **Virtual circuit:** A logical communication path established by the OSI network layer for sending and receiving data.

Layer 4: Transport Layer

- Ensures the reliability of packet transmissions from node to node, guarantees that data packets are sent and received in the same order, provides acknowledgment upon receipt, and monitors for errors to resend any bad packets.
- **Flow control:** Used to make sure one device does not send information faster than can be received by another device.

Layer 5: Session Layer

- Initiates the communication link.
- Makes sure the communication link is maintained.
- Determines which node transmits at any point in time.
- Disconnects when a communication session is over.
- Translates node addresses.

Layer 6: Presentation Layer

- Translates data to a format the receiving node understands, such as from EBCDIC to ASCII.
- Performs data encryption.
- Performs data compression.
- **EBCDIC:** Used mainly on IBM mainframe computers; specially coded 256-character set.
- **ASCII:** Commonly used character set; 96 uppercase & lowercase characters and numbers, plus 32 nonprinting characters.

Layer 7: Application Layer

- Enables sharing remote drives and printers.
- Handles e-mail messages.

- Provides file transfer, file management, and terminal emulation services.
- **Microsoft redirector:** A service used via the application layer that makes one computer visible to another for access through the network.

Communication Between Stacks

- **Building and receiving a frame:** Information moves from layer to layer.
- **Peer protocols:** Enable an OSI layer on a sending node to communicate with the same layer on a receiving node.
- **Primitive:** A command used to transfer information from one layer in an OSI stack to another layer.
- **Protocol data unit (PDU):** The information transferred between layers in the same OSI stack.
- **Service data unit (SDU):** The PDU minus control and transfer information used to transfer the data from the previous layer in the OSI stack.

DYNAMIC HOST CONFIGURATION PROTOCOL

Dynamic Host Configuration Protocol (DHCP)

- **DHCP** is a network management protocol used on Internet Protocol (IP) networks.
- It automatically assigns IP addresses and other communication parameters to devices connected to the network using a client-server architecture.
- **Why use DHCP?**
 - Every device on a TCP/IP-based network must have a unique unicast IP address to access the network and its resources.
 - Without DHCP, IP addresses for new computers or computers moved from one subnet to another must be configured manually.

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- IP addresses for computers removed from the network must be manually reclaimed.

DHCP Servers and Databases

- The network administrator establishes DHCP servers that maintain TCP/IP configuration information and provide address configuration to DHCP-enabled clients in the form of a lease offer.
- The DHCP server database includes:
 - Valid TCP/IP configuration parameters for all clients on the network.
 - Valid IP addresses, maintained in a pool for assignment to clients, as well as excluded addresses.
 - Reserved IP addresses associated with particular DHCP clients for consistent assignment.
 - The lease duration, or the length of time for which the IP address can be used before a lease renewal is required.

DHCP Lease Offer

- A DHCP-enabled client, upon accepting a lease offer, receives:
 - A valid IP address for the subnet to which it is connecting.
 - Requested DHCP options (Router/default gateway, DNS Servers, and DNS Domain Name).

How DHCP Works

- **Client-Server Model:** Consists of a centrally installed network DHCP server and client instances on each device.
- **Client Requests:** When a device connects to the network, it requests parameters (e.g., IP address) from the DHCP server.
- **Server Response:** The DHCP server provides the requested parameters to the client.

- **Dynamic IP Configuration:** Eliminates manual configuration of network devices.

Scopes and IP Address Pools

- **Scopes:** Determine which IP addresses are provided to clients.
- **DHCP Scope:** A pool of IP addresses on a subnet that the DHCP server can lease to clients.
- **Lease Duration:** The period the DHCP server holds a leased IP address for a client.

Scope Properties

- **Network ID:** Defines the range of IP addresses.
- **Subnet Mask:** Specifies the subnet for the network.
- **Router Option:** Allows DHCP clients to access remote networks.

Security Conditions

- DHCP can be secured to prevent unauthorized devices from obtaining IP addresses.
- Use DHCP snooping and port security to enhance security.

Benefits of DHCP

- **Reliable IP address configuration:** Minimizes configuration errors (typographical errors or address conflicts) caused by manual configuration.
- **Reduced network administration:**
 - Centralized and automated TCP/IP configuration.
 - Ability to define TCP/IP configurations from a central location.
 - Ability to assign a full range of additional TCP/IP configuration values by means of DHCP options.

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- Efficient handling of IP address changes for clients that must be updated frequently.
- Forwarding of initial DHCP messages by using a DHCP relay agent, eliminating the need for a DHCP server on every subnet.

SUBNETTING AND VLSM

Introduction to Subnetting

- **Definition:** Subnetting is the process of dividing a large network into smaller, more manageable subnetworks.
- **Purpose:** Improve network performance and security, efficient IP address management.

Why Subnetting is Important

- **Network Efficiency:** Reduces broadcast traffic.
- **Enhanced Security:** Limits the spread of network issues.
- **IP Address Management:** Prevents IP address wastage.

Basic Concepts

- **IP Address:** 32-bit number, divided into four octets.
- **Subnet Mask:** Determines the network and host portion of an IP address.
- **CIDR Notation:** Representation of an IP address and its associated network mask (e.g., 192.168.1.0/24).
- **CIDR "Sayder" value:** /16
- **CIDR** stands for Classless Inter-Domain Routing
- **CIDR** means subnetting

How Subnetting Works

- **Example:** Dividing a /24 network into smaller /26 networks.
- **Binary Representation:** Explaining the role of bits in subnetting.

- **Subnet Calculation:** How to calculate the number of subnets and hosts per subnet.

Class C Subnet Table

Subnet	1	2	4	8	16	32	64	128	256
Host	256	128	64	32	16	8	4	2	1
Subnet Mask	/24	/25	/26	/27	/28	/29	/30	/31	/32

Subnetting Tools and Techniques

- **Subnet Calculators:** Online tools to simplify subnetting.
- **Command-Line Tools:** Using commands like "ipcalc" on Linux.
- **Subnetting Cheat Sheets:** Quick reference guides.

Common Subnetting Mistakes

- **Overlapping Subnets:** Ensuring subnets do not overlap.
- **Incorrect Subnet Masks:** Verifying the correct subnet mask.
- **Misconfigured Router Interfaces:** Properly configuring interfaces.

Subnetting in Practice

- **Real-World Example:** Designing a network for a medium-sized office.
- **Steps:** Planning, implementation, verification.

VLSM (Variable Length Subnet Mask)

- **Definition:** Allows subnets of different sizes.
- **Benefits:** Efficient use of IP addresses.
- **Example:** Allocating IP ranges based on specific subnet size requirements.
- **VLSM Subnetting** is considered subnetting a subnet. Compared with FLSM (Fixed Length Subnet Mask) subnetting
- **VLSM subnetting** makes it more efficient of available IDs.

Steps in VLSM Subnetting

1. Arrange the networks
2. Choose the largest network and do subnetting table
3. Continue subnetting table on the next target network up to the last network
 - a. From step 2 to the last step, the process is repeating. And on the second row of the subnetting table, we are looking for the minimum number of host IDs to satisfy the given network

VLAN (VIRTUAL LOCAL AREA NETWORK)

Understanding VLAN

- A virtual local area network is a **logical subnetwork** that groups a collection of devices from different physical LANs.
- **Flat Network vs. VLANs:**
 - In a **flat network**, all devices are connected to a single switch or hub in the same broadcast area and communicate with each other without any intermediaries.
 - **VLANs** get the same result as Flat Network, at **almost half the cost** and also acquire the flexibility and expandability for the future growth of the network, which was very limited in Flat Network.
- **VLAN = Network = Subnet**
- For those LAN port available on the switch will be open for public and will request an IP address from the DHCP Server to communicate with the other devices connected to the same VLAN.

Default and Native VLAN

- A default VLAN is a VLAN that all switch ports belong to when they are not assigned to another VLAN.
- The default VLAN is usually **VLAN 1**, but it can be changed to any other number.

- It is a security best practice to configure all the ports on all switches to be associated with VLANs other than VLAN 1. This can prevent unauthorized access and broadcast storms.

Default VLAN: VLAN 1 Important points:

1. We cannot change the default VLAN, and we cannot even delete the default VLAN. It is in the default settings which is the VLAN 1.
2. VLAN 1 is never intended to be used as a standard data VLAN.

Native VLAN

- A native VLAN is a VLAN whose traffic traverses a trunk port **without a VLAN tag**.
- It is used to support legacy devices or devices that don't tag their traffic.
- It is defined in **802.1Q standard** and can be configured on both routers and switches.
- A native VLAN is usually the default VLAN on a trunk port, but it can be changed.

Why Native VLAN?

- Native VLAN is one concept defined in IEEE 802.1Q standard protocol that was created for **backward compatibility with old devices** that don't support VLANs. On an Ethernet network, all devices on the link must still be capable of communication even if they do not speak the IEEE 802.1Q standard protocol.
- The native VLAN is used by the switch to carry specific control and management protocol traffic like CISCO Discovery Protocol (CDP), VLAN Trunking Protocol (VTP), Spanning Tree Protocol (STP) or some other network management traffic does not need any VLAN tag to cross over the trunk. The idea behind the native VLAN is that some network traffic still flows over the whole switched network even when trunk or trunking fails.

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- The native VLAN is also useful when we deal with **voice over IP or VoIP**.
 - *Example of Native VLAN:* Instead of expensive and too much wirings, use this instead as the phone serves as the trunk while the PC and phone shares the same connection.

VLAN Management - Create

- Commands on how to create a VLAN on a switch:

```
#en
#conf t
#vlan 300
#name SALES
#exit
```

- To verify the VLAN use this command:

```
#sh vlan brief
```

VLAN Management – IP config

- Setting up an IP, default subnet mask of the created VLAN:

```
#interface vlan 300
#ip address 192.168.0.1 255.255.255.0
#exit
```

- Set the trunk based on the FastEthernet connected port:

```
#interface fastEthernet 4/1
#switchport mode trunk
```

VLAN Management – Checking

- Check all the list of active VLANs: `#show vlan`
- Check all the status of the VLANs: `#show run`
- Check the connections of the VLANs: `#ping 192.168.0.2`
- Show some suggestion of commands: `#?`

- Set a Username and Password on the Switch: `(conf t)#username abc password 123`

VLAN Management – Delete

- Command to delete the VLAN:

```
#sh vlan
#erase start
#reload
#en
#sh vlan
#delete vlan.dat
```

Assigning port to VLAN

```
#en
#conf t
#int fastEthernet 4/2
#switchport access vlan 10
#exit
#int fastEthernet 4/3
#switchport access vlan 20
#exit
#int fastEthernet 4/4
#switchport access vlan 30
#exit
```

Inter-VLAN Routing

- 3 Options of Inter-VLAN Routing:
 1. **Traditional**
 - a. The traditional Inter-VLAN uses a router as a mediator for data transfer from 1 VLAN to another VLAN.
 - b. But the number of physical interfaces on a router are limited and expensive.
 - c. This method is applicable to 2 or 3 VLANs.
 2. **Router-on-a-stick**
 - a. The router must understand the concept of VLAN and the IEEE 802.1Q standard.

- b. With Router-on-a-stick Inter VLAN Routing, we configure **one single link as a trunk**.

3. Multilayer Switch

- a. Multilayer Switch Inter-VLAN or Layer-3 switch, switching and routing are achieved inside one box.
- b. It supports **Switch Virtual Interfaces or SVIs** that act as gateways and perform routing. One SVI can be created for each VLAN.

NETWORK SECURITY

What is Network Security?

- **Network security** is the protection of the underlying networking infrastructure from unauthorized access, misuse, or theft.
- It involves creating a secure infrastructure for devices, applications, users, and applications to work in a secure manner.

How does Network Security Work?

- Network security combines multiple layers of defenses at the edge and in the network.
- Each network security layer implements policies and controls.
- Authorized users gain access to network resources, but malicious actors are blocked from carrying out exploits and threats.
- Worse yet, increasingly sophisticated cybercriminals are exploiting network vulnerabilities at an alarming rate.
- **Malware, ransomware, distributed denial-of-service (DDoS) attacks**, and countless other threats are challenging IT teams to fortify their defenses.

Benefits of Network Security

- **Reduced cyber risk:** Strong, robust security measures help ensure your data always stays protected.

- **Enhanced data privacy:** Eliminating threat vectors ensures sensitive information is shielded from unauthorized access while traversing the network, thereby safeguarding customer data, and maintaining compliance.
- **Improved business continuity:** Protected networks are more resilient against potential disruptions and experience minimal downtime, leading to optimal productivity.
- **Better network performance:** Security prevents bad actors from disabling the network, ensuring critical resources are always readily available.

Why do enterprises need Network Security?

- Digitization has transformed our world. How we live, work, play, and learn have all changed.
- Every organization that wants to deliver the services that customers and employees demand must protect its network.
- Network security also helps you protect proprietary information from attack. Ultimately it protects your reputation.
- Vulnerabilities in your network security give hackers an opening to do untold damage to your network while exposing potentially sensitive and confidential information.
- Network security protection has been developed to implement measures to protect your computer network's data from being lost, stolen, or manipulated.

Essential Devices for Securing Network Infrastructure

- **Ethernet switches:** Switching ensures adequate security at the network edge by facilitating traffic filtering and access control at the port level, allowing administrators to enforce policies on granular network segments.
- **Wi-Fi access points (AP):** Wireless APs implement encryption protocols and

authentication mechanisms, protecting data in transit. They also support access control lists (ACLs), restricting unauthorized devices from connecting to the network.

- **Gateways:** 5G and LTE gateways are key to redundant and primary links to branch offices and campuses alike. Housing these devices under the same security umbrella as the rest of the network ensures common configuration, thereby reducing the attack surface.

Types of Network Security Solutions

1. Firewalls

- a. A firewall is a device that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules.
- b. Acting as a barrier between trusted internal and untrusted external networks, it works by inspecting data packets and choosing to block or allow them.

2. Intrusion Prevention Systems (IPS)

- a. Intrusion prevention systems detect, and block known and suspected threats before they can impact the network core or devices at its edge.
- b. Using an IPS, organizations can rapidly detect attack signatures and abnormal behavior.
- c. The system automatically takes action to block malicious traffic while alerting administrators for further investigation.

3. Antivirus and Sandboxing

- a. Antivirus and sandboxing tools are key to determining whether a file is malicious.
- b. While antivirus blocks known malware threats, sandboxing provides a safe environment to analyze suspicious files.

4. Web and DNS Filtering

- a. Domain Name System (DNS) filtering allows organizations to stop

domain-based attacks, such as DNS hijacking, tunneling, etc.

- b. Likewise, URL filtering prevents users and applications from accessing suspicious URLs, which could be linked to malicious websites.

5. Remote Access VPNs

- a. Remote access VPNs allow users to securely access the corporate network from outside their organization's office.
- b. They create a private, encrypted connection from a public Wi-Fi network, enabling employees to safely use critical resources from their personal devices regardless of location.

6. Network Access Control (NAC)

- a. is a security solution that uses a set of protocols to keep unauthorized users and devices out of a private network or give restricted access to the devices which are compliant with network security policies.
- b. It is also known as Network Admission Control.
- c. It handles network management and security that implements security policy, compliance, and management of access control to a network.